

# Agent Harness vs. MCP Server (One-Page Summary)

## Agent Harness

An agent harness is the runtime layer around an LLM. It manages the agent loop, tool execution, memory, planning, context, safety, retries, and observability. It decides **when** a tool should be used and coordinates the workflow.

## MCP (Model Context Protocol) Server

An MCP server exposes tools, data, and resources through a standard protocol that AI agents can use. It performs the requested operations (e.g., search Jira, query a database, read files, or call APIs). It defines **what** capabilities are available.

## How They Work Together

User → LLM → Agent Harness → MCP Client → MCP Server → API/Database → Results → Harness → LLM → User

## Responsibilities

- LLM: Reasons about the request.
- Agent Harness: Orchestrates execution, memory, safety, retries, and tool calls.
- MCP Server: Provides standardized tools and resources.
- Application/API: Performs the actual business logic.

## Example (AskNSDI)

User asks for wildfire datasets. The harness determines a search is needed, calls the GeoPlatform MCP server, which queries Solr and returns matching datasets. The harness sends the results back to the LLM, which generates the final response.